

Your grade: 94.28%

Your latest: 94.28% • Your highest: 94.28% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Calculate the value of the following limit:

1 / 1 point

$$\lim_{x \rightarrow 2} f(x) = \frac{x^2 + 4x - 12}{x^2 - 2x}.$$

If the correct answer is $n = 1$, then insert only the value 1, or if the answer is $n = -1$, then insert -1 only without any spaces.

4

👍 Nice work

2. Consider the function

0.7 / 1 point

$$f(x) = \begin{cases} x^3 + 2, & \text{if } x < 2 \\ 5 & \text{if } x = 2 \\ x^2 + 6, & \text{if } x > 2 \end{cases}$$

Select all the correct statements.

- ☐ The limit of the function is equal to 10
- ☐ The function value is equal to 5
- ☒ the limit exists at $x = 2$

👍 Nice work

- ☒ it is discontinuous at $x = 2$

👍 Nice work

- ☐ The function value is equal to 2
- ☐ the limit does not exists at $x = 2$
- ☐ None of the other options is correct

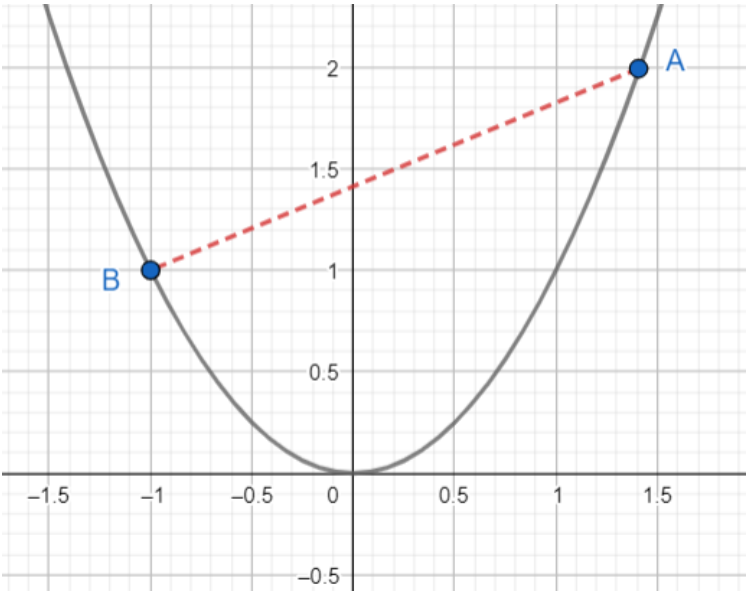
⊗ You didn't select all the correct answers

3. The graph below represents the function $f(x) = x^2$.

1 / 1 point

Write the coordinates of the points A and B .

Notice the chord between A and B .



Calculate the gradient of the chord AB .

Select all correct statements.

- ☒ A has the coordinates $(1.4, 2)$

👍 Nice work

- ☐ A has the coordinates $(2, 1.4)$
- ☐ B has the coordinates $(1, -1)$
- ☒ B has the coordinates $(-1, 1)$

👍 Nice work

- ☒ The gradient of the chord AB is $\frac{1}{2.4}$

👍 Nice work

- ☐ The gradient of the chord AB is 2.4

4. If $f(x) = \frac{1}{x}$ and $g(x) = x^2 + 2$ find the composite function $f \circ g(x)$.

1 / 1 point

Select all the correct answers

- ☒ $f(g(x)) = \frac{1}{x^2+2}$

👍 Nice work

- ☐ $f(g(x)) = x^2 + 2$
- ☐ $f(g(x)) = \frac{1}{x}$
- ☐ $f(g(x)) = x + \frac{2}{x}$
- ☒ $f(g(x)) = \frac{x}{x^2+2x}$

👍 Nice work

5. Find the local maxima and minima points for

1 / 1 point

$$y = x^4 - 2x^2 + 3$$

- ☒ Local maximum: (0,3)

Local minima: (-1,2)(1,2)

- ☐ Local maximum: (0,0)

Local minima: (-1,2)(1,2)

- ☐ Local maximum: (0,0)

Local minima: (1,2)(1,2)

- ☐ min at $x = 0$, max at $x = 0$

- ☐ None of the other options is correct

👍 Nice work